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SCHOOL OF NATURAL SCIENCES
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Designing a Modular Online Learning System

PROBLEM STATEMENT

Design an Online Learning Platform with these features:

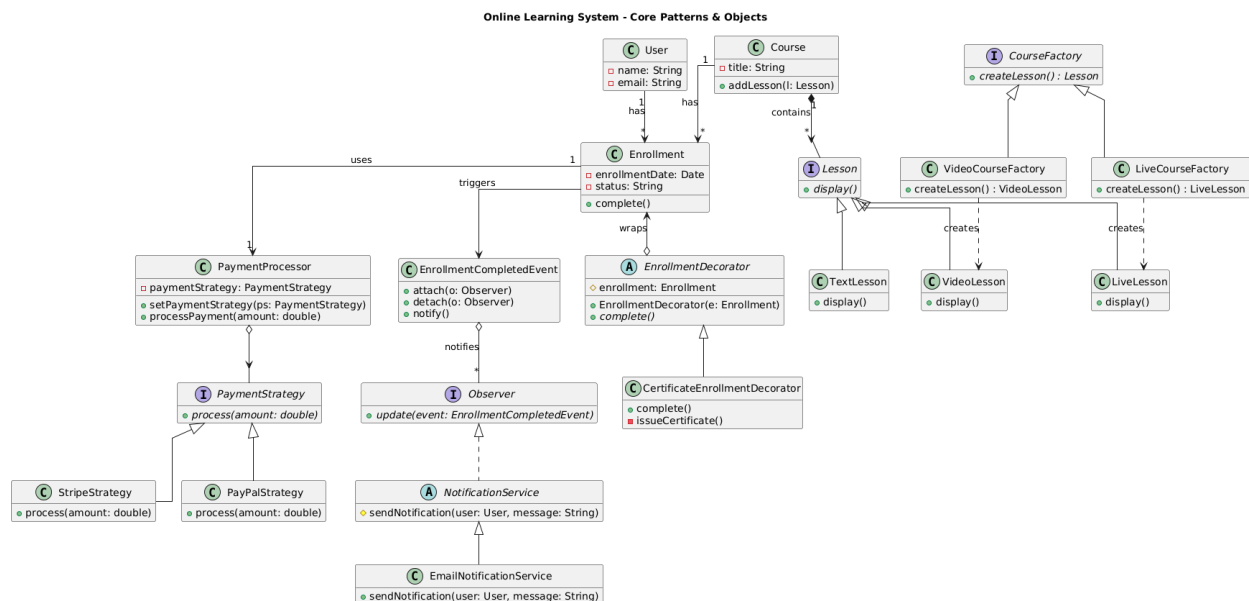
1. Multiple Course Formats: Video, Text, Live (e.g., Zoom-based).
2. Flexible Payment Processing: Support multiple payment gateways.
3. User Notifications: Alerts on enrollment/completion.
4. Dynamic Course Enhancements: Add features like certificates/badges post-enrollment.

Deliverables

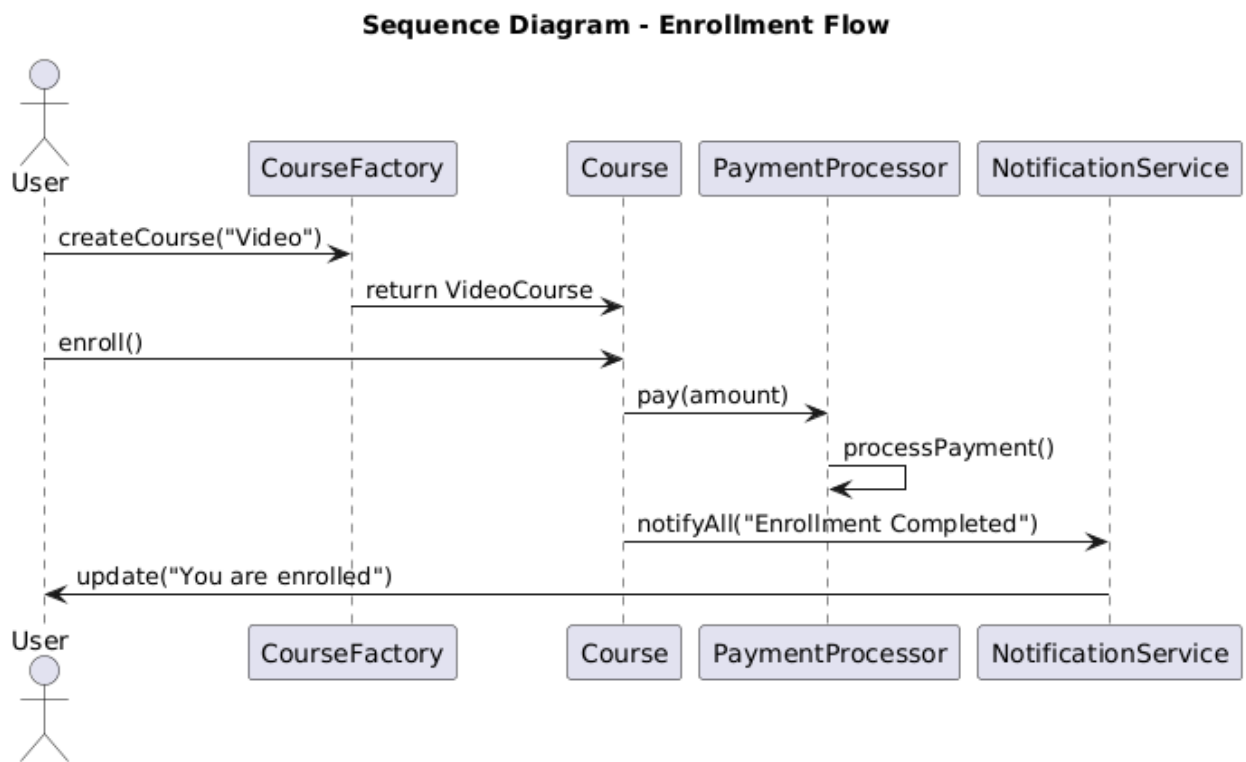
Core objects

- User
- Course
- Enrollment
- PaymentProcessor
- NotificationService

1. CLASS UML DIAGRAM



2. SEQUENCE DIAGRAM



Pain Points & Pattern Triggers

Pain Point	Why it's a Problem	Pattern(s) to Consider
Course Instantiation	Different course formats (Video, Text, Live)	Factory Method or Abstract Factory
Payment Processing	Multiple gateways needed (e.g., PayPal, Stripe)	Strategy
User Notifications	Must decouple alerts (e.g., Email, SMS)	Observer
Add Certificates/Badges at Runtime	Post-enrollment features, avoid subclass bloat	Decorator

Pattern Discovery & Application

Pattern 1: Factory Method

Used for: Course creation (video/text/live).

Justification: Creation logic varies; applying Creator and Low Coupling.

Pattern 2: Strategy

Used for: Payment gateway switching.

Justification: Behavior varies independently; Polymorphism, Low Coupling.

Pattern 3: Observer

Used for: Sending notifications on events (enrollment/completion).

Justification: Indirection and Low Coupling avoid hardwiring all outputs.

Pattern 4: Decorator

Used for: Dynamically adding certificates or badges.

Justification: Avoid subclassing every feature combination. Use High Cohesion, Open-Closed Principle.